1a. Oxygen demand D^{Oxy} (unknown $f = \log_{10} f_{\mathrm{O_2}}$)

species	excess O	why
FeO, H_2O, C (graphite, diamond)	0	the reference
Fe^{3+} ($FeO_{1.5}$), solid and melt	$+\frac{1}{2}$ per Fe	FeO $+\frac{1}{2}$O
Fe metal	-1 per Fe	FeO $-$ O, the most reduced the solid can be
H_2, fluid and melt	-1 per H_2	H_2O $-$ O
CH_4	-2 per C	C + $2H_2$, two waters without their O
CO	$+1$ per C	C + O
CO_2, fluid and melt	$+2$ per C	C + 2O
carbonate (CO_3^{2-} with MgO, CaO)	$+2$ per C	as CO_2, the oxide part is neutral

What is solved. The cell holds an excess-oxygen budget of TOex M grams (TOex per gram of cell, carried by the tracers; M the cell mass). Each sink of the cell holds an amount of this oxygen that grows with the cell's fO_2. The *oxygen demand* $D^{\mathrm{Oxy}}(f)$ is what all sinks together hold at a trial f, and the solve looks for the f at which it equals the budget:

$$F^{\mathrm{Oxy}}(f) = D^{\mathrm{Oxy}}(f) - \mathrm{TOex}\, M = 0 \tag{1}$$

$$D^{\mathrm{Oxy}}(f) = \underbrace{s_{\mathrm{ex}}}_{\text{solid } Fe^{3+}} + \underbrace{m_{\mathrm{ex}}}_{\text{melt } Fe^{3+}} + \underbrace{\Phi \frac{X^m_{\mathrm{CO_2}}}{1 - X^m_{\mathrm{CO_2}}}}_{\text{melt } CO_2} + \underbrace{2 m_{\mathrm{O}}\, n^{\mathrm{carb}}_{\mathrm{C}}}_{\text{carbonate}} + \underbrace{m_{\mathrm{O}}\, n \big(2X_{\mathrm{CO_2}} + X_{\mathrm{CO}} - 2X_{\mathrm{CH_4}} - X_{\mathrm{H_2}}\big)}_{\text{fluid}} \underbrace{- m_{\mathrm{O}}\, n^m_{\mathrm{H_2}}}_{\text{melt } H_2} \tag{2}$$

with every term in grams of O and m_{O} the molar mass of O. The other sinks are introduced in Secs. 2–4. Every sink holds more oxygen at higher f, so D^{Oxy} rises with f.

1b. Oxygen: the iron sinks

Both iron sinks are read at the trial f by inverting an oxybarometer:

- **Solid Fe^{3+}**, s_{ex} (g of O). MAGEMin gives the fO_2 of the solid, $f_s(s)$, as a function of its ferric load s. The solve inverts it at the trial f:

$$f_s(s_{ex}) = f, \qquad -\underbrace{s_{floor}}_{\text{all metal}} \le s_{ex} \le \underbrace{s_{stoich}}_{\text{all ferric}} = \tfrac{1}{2}\, n^s_{Fe}\, m_O$$

  with n^s_{Fe} the moles of Fe in the solid. Above the MAGEMin table the solid's fO_2 is extrapolated in log, with $r = s/s_{stoich}$ its $Fe^{3+}/\Sigma Fe$:

$$f_s(s) = f_N + \frac{1}{0.1917}\Big[\log_{10}\frac{r}{1-r} - \log_{10}\frac{R_{max}}{1-R_{max}}\Big], \qquad r > R_{max}$$

  with $R_{max} = 0.4$ the top of the MAGEMin table and f_N its fO_2 there.

- **Melt Fe^{3+}**, m_{ex} (g of O). Hirschmann (2022) gives f from the melt's Fe^{3+}/Fe^{2+} ratio ρ; the solve inverts it at the trial f:

$$f = \frac{1}{a}\Big[\log_{10}\rho - b - \frac{c}{T} + \frac{\Delta C_p}{R\ln 10}\Big(1 - \frac{T_0}{T} - \ln\frac{T}{T_0}\Big) + \frac{\int_1^P \Delta V\, dP}{RT\ln 10} - \frac{Y}{T}\Big], \qquad \rho = \frac{2m_{ex}/m_O}{n^m_{Fe} - 2m_{ex}/m_O}$$

$$Y = y_1 X_{Si} + y_3 X_{Mg} + y_4 X_{Ca} + y_5 X_{Na} + y_8 X_{Si} X_{Al} + y_9 X_{Si} X_{Mg}$$

  with a, b, c, ΔC_p, T_0, y_k the constants of the law, Y its composition term, X_k the cation fractions of the melt and n^m_{Fe} its moles of Fe.

1c. Oxygen: variables

variable	what it is
$f = \log_{10} f_{O_2}$	oxygen fugacity of the cell, the unknown
$D^{\mathrm{Oxy}}(f)$	oxygen demand: excess O all sinks hold at f, g
$F^{\mathrm{Oxy}}(f)$	oxygen residual, demand minus budget, g
TOex, M	excess-oxygen budget per gram of cell (tracers), and cell mass (1000 g)
m_{O}	molar mass of oxygen
s_{ex}, m_{ex}	excess oxygen held as Fe^{3+} in the solid and in the melt
$f_s(s)$	fO_2 of the solid at ferric load s (MAGEMin sb24 table)
$R_{\max}$, f_N	top of the MAGEMin table in $Fe^{3+}/\Sigma Fe$ (0.4), and the fO_2 there
$r = s/s_{\mathrm{stoich}}$	ferric fraction $Fe^{3+}/\Sigma Fe$ of the solid
s_{floor}, s_{stoich}	solid load with all Fe as metal, and with all Fe ferric
n^s_{Fe}, n^m_{Fe}	moles of Fe in the solid and in the melt
ρ	Fe^{3+}/Fe^{2+} of the melt

variable	what it is
a, b, c	constants of the melt oxybarometer (Hirschmann 2022)
ΔC_p, T_0	its heat-capacity term and reference temperature
$y_1 \dots y_9$	its melt-composition coefficients
$\int \Delta V \, dP$	its volume term, per P and T
$X_{\mathrm{Si}} \dots X_{\mathrm{Na}}$	cation fractions of the basaltic melt
$\Phi = 2 m_{\mathrm{O}} n_{\mathrm{ox}}$	grams of O per unit $X/(1-X)$ of melt CO_2
n_{ox}	moles of oxide in the melt
$X^m_{\mathrm{CO_2}}$	CO_2 mole fraction of the melt (Sec. 3)
$n^{\mathrm{carb}}_{\mathrm{C}}$	moles of carbon in carbonate (Sec. 3)
n, X_i	fluid moles and species mole fractions (Sec. 2)
$n^m_{\mathrm{H_2}}$	moles of H_2 dissolved in the melt (Sec. 4)
P, T	pressure and temperature of the cell

2a. Fluid closure s_X^{Flu} (unknown n, the fluid moles, solved as $u = \ln n$)

The species. At the trial fO_2 the four C–O–H equilibria

$$\begin{aligned} \mathrm{C} + \mathrm{O_2} &= \mathrm{CO_2}, & K_1 &= \frac{f_{\mathrm{CO_2}}}{a_{\mathrm{C}}\, f_{\mathrm{O_2}}} \\ \mathrm{C} + \tfrac{1}{2}\mathrm{O_2} &= \mathrm{CO}, & K_2 &= \frac{f_{\mathrm{CO}}}{a_{\mathrm{C}}\, f_{\mathrm{O_2}}^{1/2}} \\ \mathrm{H_2} + \tfrac{1}{2}\mathrm{O_2} &= \mathrm{H_2O}, & K_3 &= \frac{f_{\mathrm{H_2O}}}{f_{\mathrm{H_2}}\, f_{\mathrm{O_2}}^{1/2}} \\ \mathrm{C} + 2\mathrm{H_2} &= \mathrm{CH_4}, & K_4 &= \frac{f_{\mathrm{CH_4}}}{a_{\mathrm{C}}\, f_{\mathrm{H_2}}^{2}} \end{aligned} \qquad K_j = \exp\!\big(-\Delta G_j^{\circ}/RT\big) \qquad f_i = \varphi_i\, X_i\, P$$

fix every fluid species from two potentials, the carbon activity $a_{\mathrm{C}} = e^{x}$ (Sec. 3) and the H_2 fugacity $f_{\mathrm{H_2}} = e^{y}$ (Sec. 4):

$$X_{\mathrm{H_2O}} = \frac{K_3 f_{\mathrm{H_2}} \sqrt{f_{\mathrm{O_2}}}}{\varphi_{\mathrm{H_2O}} P}, \quad X_{\mathrm{H_2}} = \frac{f_{\mathrm{H_2}}}{\varphi_{\mathrm{H_2}} P}, \quad X_{\mathrm{CO_2}} = \frac{a_{\mathrm{C}} K_1 f_{\mathrm{O_2}}}{\varphi_{\mathrm{CO_2}} P}, \quad X_{\mathrm{CO}} = \frac{a_{\mathrm{C}} K_2 \sqrt{f_{\mathrm{O_2}}}}{\varphi_{\mathrm{CO}} P}, \quad X_{\mathrm{CH_4}} = \frac{a_{\mathrm{C}} K_4 f_{\mathrm{H_2}}^2}{\varphi_{\mathrm{CH_4}} P} \tag{3}$$

f_i is the fugacity of species i and K_j the equilibrium constant of reaction j, from its standard Gibbs energy ΔG_j° with carbon at its stable polymorph, $\mu_{\mathrm{C}}^{\circ} = \min(G_{\mathrm{gph}}, G_{\mathrm{dia}})(P,T)$, and the gases at 1 bar (HP98). φ_i is the fugacity coefficient of species i (Sec. 6), held fixed during this solve, and P, T are the cell's pressure and temperature.

The closure. A fluid can exist only if its mole fractions add up to one. Their sum is s_X^{Flu}:

$$s_X^{\mathrm{Flu}} = \sum_i X_i, \qquad F^{\mathrm{Flu}}(u) = \ln s_X^{\mathrm{Flu}}\big(x(u),\, y(u)\big) = 0, \qquad n \geq 0, \quad s_X^{\mathrm{Flu}} \leq 1, \quad n\,(1 - s_X^{\mathrm{Flu}}) = 0 \tag{4}$$

It depends on n only indirectly. The carbon and hydrogen closures (Secs. 3–4) share their budgets between the fluid and the other sinks, so their potentials $x(u)$ and $y(u)$ depend on how much fluid there is. More fluid moles leave less of each element per mole, the potentials drop, and s_X^{Flu} falls with u.

The switch. Whether a fluid exists is decided first. Carbon and hydrogen are solved at $n = 0$, and if $s_X^{\mathrm{Flu}} \leq 1$ there is no fluid. Only otherwise is u solved, so no step ever crosses the switch. Above 50 GPa the fluid is taken as ideal pure water.

2b. Fluid: the derivative

Species. Each species is a power law in the two potentials, $X_i \propto a_{\mathrm{C}}^{\nu_{C,i}} f_{\mathrm{H_2}}^{\nu_{H,i}}$, with exponents (ν_C, ν_H) of $(0,1)$ for H_2O and H_2, $(1,0)$ for CO_2 and CO, and $(1,2)$ for CH_4. So

$$\frac{\partial X_i}{\partial x} = \nu_{C,i} X_i, \qquad \frac{\partial X_i}{\partial y} = \nu_{H,i} X_i, \qquad \frac{\partial s_X^{\mathrm{Flu}}}{\partial x} = X_{\mathrm{CO_2}} + X_{\mathrm{CO}} + X_{\mathrm{CH_4}}, \qquad \frac{\partial s_X^{\mathrm{Flu}}}{\partial y} = X_{\mathrm{H_2O}} + 2X_{\mathrm{CH_4}} + X_{\mathrm{H_2}} \tag{5}$$

How x and y move with u. The carbon and hydrogen residuals $F^{\mathrm{Car}}(x,y,u)$ and $F^{\mathrm{Hyd}}(x,y,u)$ (Secs. 3–4) stay zero as u changes, so the direct effect of u is cancelled by the shift of the roots:

$$\frac{\partial F^{\mathrm{Car}}}{\partial u} + \frac{\partial F^{\mathrm{Car}}}{\partial x}\frac{dx}{du} + \frac{\partial F^{\mathrm{Car}}}{\partial y}\frac{dy}{du} = 0, \qquad \frac{\partial F^{\mathrm{Hyd}}}{\partial u} + \frac{\partial F^{\mathrm{Hyd}}}{\partial x}\frac{dx}{du} + \frac{\partial F^{\mathrm{Hyd}}}{\partial y}\frac{dy}{du} = 0 \tag{6}$$

or, as one matrix equation in the two unknown shifts,

$$\underbrace{\begin{pmatrix} \partial F^{\mathrm{Car}}/\partial x & \partial F^{\mathrm{Car}}/\partial y \\ \partial F^{\mathrm{Hyd}}/\partial x & \partial F^{\mathrm{Hyd}}/\partial y \end{pmatrix}}_{\mathbf{J}} \begin{pmatrix} dx/du \\ dy/du \end{pmatrix} = - \begin{pmatrix} \partial F^{\mathrm{Car}}/\partial u \\ \partial F^{\mathrm{Hyd}}/\partial u \end{pmatrix} \tag{7}$$

so their roots move as

$$\begin{pmatrix} dx/du \\ dy/du \end{pmatrix} = -\mathbf{J}^{-1} \begin{pmatrix} \partial F^{\mathrm{Car}}/\partial u \\ \partial F^{\mathrm{Hyd}}/\partial u \end{pmatrix} \tag{8}$$

where $\mathbf{J}$ is the Jacobian of the two closures, whose entries are derived in 3b and 4b.

The derivative of the fluid residual is then the chain rule through both potentials:

$$\frac{dF^{\mathrm{Flu}}}{du} = \frac{1}{s_X^{\mathrm{Flu}}}\Big(\frac{\partial s_X^{\mathrm{Flu}}}{\partial x}\frac{dx}{du} + \frac{\partial s_X^{\mathrm{Flu}}}{\partial y}\frac{dy}{du}\Big) < 0 \tag{9}$$

Newton on u.

$$u_{k+1} = u_k - \frac{F^{\mathrm{Flu}}(u_k)}{dF^{\mathrm{Flu}}/du}, \qquad u \in \big[\ln n_{\max} - 70,\ \ln n_{\max}\big], \qquad n_{\max} = N_{\mathrm{C}} + N_{\mathrm{H}}/2 \tag{10}$$

$n_{\max}$ is the fluid of a cell whose whole carbon budget N_{C} and hydrogen budget N_{H} (moles) were in the fluid. A step leaving the bracket is replaced by bisection. Tolerance $|F^{\mathrm{Flu}}| \le 10^{-12}$.

2c. Fluid: variables

variable	what it is
$u = \ln n$	log of the fluid moles, the unknown
n	moles of fluid in the cell
s_X^{Flu}	sum of the species mole fractions
$F^{\mathrm{Flu}} = \ln s_X^{\mathrm{Flu}}$	fluid residual
X_i	mole fraction of species i (H_2O, H_2, CO_2, CO, CH_4)
$a_{\mathrm{C}} = e^x$	carbon activity (Sec. 3)
$f_{\mathrm{H_2}} = e^y$	H_2 fugacity (Sec. 4)
$f_{\mathrm{O_2}}$	trial fO_2 of the outer solve (Sec. 1)
$\nu_{C,i}$, $\nu_{H,i}$	exponents of a_{C} and $f_{\mathrm{H_2}}$ in X_i
$n_{\max}$	upper end of the bracket of n

variable	what it is
$K_1 \ldots K_4$	equilibrium constants of the four reactions (HP98)
ΔG_j°	standard Gibbs energy of reaction j
μ_{C}°	standard potential of carbon, graphite or diamond
φ_i	fugacity coefficient of species i (Sec. 6), fixed here
P, T	pressure and temperature of the cell
F^{Car}, F^{Hyd}	carbon and hydrogen residuals (Secs. 3, 4)
$\mathbf{J}$	Jacobian of $(F^{\mathrm{Car}}, F^{\mathrm{Hyd}})$ in (x, y)
N_{C}, N_{H}	carbon and hydrogen budgets, moles

3a. Carbon demand D^{Car} (unknown $x = \ln a_{\mathrm{C}}$)

What is solved. The cell holds N_{C} moles of carbon (tracers). It can sit dissolved as CO_2 in the melt, in the fluid, or as a solid (graphite, diamond or carbonate), and how much each sink holds grows with the carbon activity a_{C}. The *carbon demand* $D^{\mathrm{Car}}(x)$ is what all three hold at a trial $x = \ln a_{\mathrm{C}}$, with the fluid amount n and the hydrogen potential y held at their trial values:

$$D^{\mathrm{Car}}(x) = \underbrace{n_{\mathrm{ox}} \frac{X^m_{\mathrm{CO_2}}}{1 - X^m_{\mathrm{CO_2}}}}_{\text{melt CO}_2} + \underbrace{n\big(X_{\mathrm{CO_2}} + X_{\mathrm{CO}} + X_{\mathrm{CH_4}}\big)}_{\text{fluid}} + \underbrace{n^{\mathrm{sol}}_{\mathrm{C}}(x)}_{\text{solid}}, \qquad F^{\mathrm{Car}} = \ln \frac{D^{\mathrm{Car}}}{N_{\mathrm{C}}} = 0 \tag{11}$$

Melt CO_2. n_{ox} is the moles of oxide in the melt and $X^m_{\mathrm{CO_2}}$ its CO_2 mole fraction. A graphite-saturated melt at fO_2 f holds B_m (Stagno & Frost 2010), and a_{C} scales it:

$$v = a_{\mathrm{C}} B_m, \qquad B_m = 10^{\,f - 5.44 + 21380/T - 0.078(P-1)/T}, \qquad X^m_{\mathrm{CO_2}} = \mathrm{softcap}\big(v,\ X_{\max}\big) \tag{12}$$

v is the melt CO_2 before any limit, and $X_{\max} = 0.5$ its ceiling. The *softcap* is a smooth minimum, close to v well below the ceiling and to $X_{\max}$ above it, so the demand has a continuous slope:

$$\mathrm{softcap}(v, c) = \max\Big(0,\ -\tfrac{1}{\beta} \ln\left(e^{-\beta v} + e^{-\beta c}\right)\Big), \qquad \beta = 30/c \quad \text{(sharpness)} \tag{13}$$

Fluid. The carbon species of Sec. 2, n moles of fluid times their mole fractions.

Solid. At saturation the solid holds whatever melt and fluid do not. Saturation is at x_{cap}, set by the EMOG(D) reaction $2MgCO_3 + Mg_2Si_2O_6 = 2Mg_2SiO_4 + 2C + 2O_2$ with standard Gibbs energy $\Delta G^\circ_{\mathrm{EMOG}}$ (HP98). Below it the solid share falls by a factor e for every $w = 0.01 \ln 10$ in $\ln a_{\mathrm{C}}$:

$$n^{\mathrm{sol}}_{\mathrm{C}}(x) = N_{\mathrm{C}} \exp\Big(\frac{\min(x - x_{\mathrm{cap}}, 0)}{w}\Big), \qquad x_{\mathrm{cap}} = \ln \min\left(1, 10^{\,f_E - f}\right), \qquad f_E = -\frac{\Delta G^\circ_{\mathrm{EMOG}}}{2RT \ln 10}$$
$$n^{\mathrm{carb}}_{\mathrm{C}} = \chi\, n^{\mathrm{sol}}_{\mathrm{C}}, \qquad n^{\mathrm{gph}}_{\mathrm{C}} = (1 - \chi)\, n^{\mathrm{sol}}_{\mathrm{C}}, \qquad \chi = \Big[1 + e^{-(f - f_E)/0.05}\Big]^{-1} \tag{14}$$

f_E is the fO_2 of EMOG(D). Above it the solid carbon is carbonate, below it graphite or diamond, and χ switches between them over 0.05 log units.

3b. Carbon: the derivative

The softcap's slope is the weight of v in the smooth minimum, 1 well below the ceiling and 0 above it (0 also where the outer max clips to 0):

$$\sigma(v) = \frac{d\,\mathrm{softcap}}{dv} = \frac{1}{1 + e^{\beta(v - X_{\max})}} \tag{15}$$

The derivative. The solve differentiates $F^{\mathrm{Car}} = \ln D^{\mathrm{Car}} - \ln N_{\mathrm{C}}$, so each entry is a partial of D^{Car} divided by D^{Car}. Since $dv/dx = v$ and $dX_i/dx = \nu_{C,i} X_i$ (2b):

$$\frac{\partial F^{\mathrm{Car}}}{\partial x} = \frac{1}{D^{\mathrm{Car}}}\Big[\underbrace{\frac{n_{\mathrm{ox}}\,\sigma(v)\,v}{(1 - X^{m}_{\mathrm{CO_2}})^2}}_{\text{melt}} + \underbrace{n\big(X_{\mathrm{CO_2}} + X_{\mathrm{CO}} + X_{\mathrm{CH_4}}\big)}_{\text{fluid}} + \underbrace{\frac{n^{\mathrm{sol}}_{\mathrm{C}}}{w}\mathbb{1}[x < x_{\mathrm{cap}}]}_{\text{solid}}\Big] > 0 \tag{16}$$

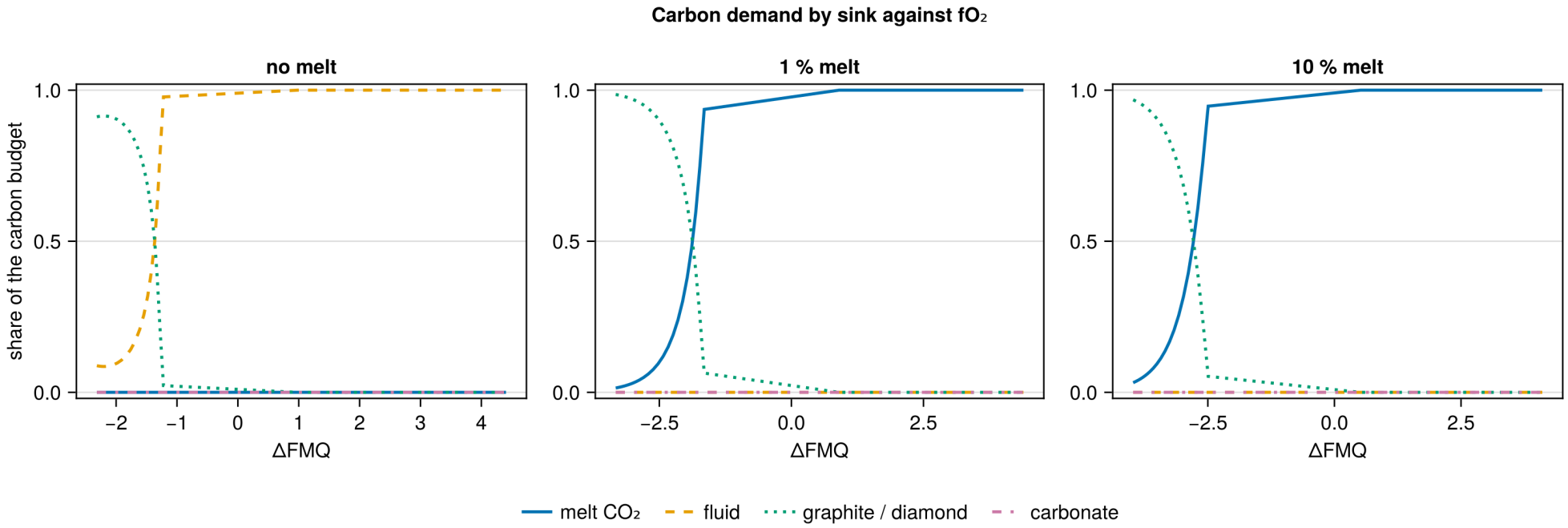
with $\mathbb{1}[\cdot]$ equal to 1 where its condition holds and 0 elsewhere. The couplings to hydrogen (only through CH_4) and to the fluid amount are

$$\frac{\partial F^{\mathrm{Car}}}{\partial y} = \frac{2n X_{\mathrm{CH_4}}}{D^{\mathrm{Car}}}, \qquad \frac{\partial F^{\mathrm{Car}}}{\partial u} = \frac{n\big(X_{\mathrm{CO_2}} + X_{\mathrm{CO}} + X_{\mathrm{CH_4}}\big)}{D^{\mathrm{Car}}} \tag{17}$$

At the cap the solid term stops growing and $\partial F^{\mathrm{Car}}/\partial x$ steps down. The cap is the upper end of the bracket $[x_{\mathrm{cap}} - 700,\ x_{\mathrm{cap}}]$ and, with graphite saturated, often the root itself. The slope is taken from below there, otherwise the Jacobian is singular. Without fluid ($n = 0$) carbon no longer depends on y and the carbon and hydrogen closures decouple.

Computed by ForwardDiff (dual numbers) in one pass with D^{Car}; the expressions above are what it returns. D^{Car} rises monotonically with x, so the root is unique. Tolerance $|F^{\mathrm{Car}}| \le 10^{-12}$.

3c. Carbon: demand by sink against fO_2



3d. Carbon: variables

variable	what it is
$x = \ln a_{\mathrm{C}}$	log of the carbon activity, the unknown
$D^{\mathrm{Car}}(x)$	carbon demand: moles of carbon all sinks hold at x
F^{Car}	carbon residual, $\ln(D^{\mathrm{Car}}/N_{\mathrm{C}})$
N_{C}	carbon budget of the cell, moles (tracers)
y, $u = \ln n$	H_2 potential and fluid amount, at their trial values
n_{ox}	moles of oxide in the melt
$X^m_{\mathrm{CO_2}}$	CO_2 mole fraction of the melt
B_m	melt CO_2 of a graphite-saturated melt at f (Stagno & Frost 2010)
$v = a_{\mathrm{C}} B_m$	melt CO_2 before its ceiling
$X_{\max}$	ceiling of the melt CO_2
softcap, β	smooth minimum and its sharpness
$\sigma(v)$	slope of the softcap

variable	what it is
$n^{\mathrm{sol}}_{\mathrm{C}}(x)$	moles of solid carbon (graphite, diamond and carbonate)
x_{cap}	$\ln a_{\mathrm{C}}$ at saturation, EMOG(D) or $a_{\mathrm{C}} = 1$
w	width of the solid ramp below the cap, in $\ln a_{\mathrm{C}}$
f_E	fO_2 of the EMOG(D) boundary
$\Delta G^{\circ}_{\mathrm{EMOG}}$	standard Gibbs energy of the EMOG(D) reaction (HP98)
χ	carbonate share of the solid carbon
$n^{\mathrm{carb}}_{\mathrm{C}}$, $n^{\mathrm{gph}}_{\mathrm{C}}$	moles of carbon in carbonate and in graphite or diamond
n, X_i	fluid moles and species fractions (Sec. 2)
$\mathbb{1}[\cdot]$	1 where the condition holds, else 0

4a. Hydrogen demand D^{Hyd} (unknown $y = \ln f_{\mathrm{H_2}}$)

What is solved. The cell holds N_{H} moles of H (tracers, $2M_{\mathrm{H_2O}}/m_{\mathrm{H_2O}}$ with $m_{\mathrm{H_2O}}$ the molar mass of water). It can sit as $\mathrm{H_2}$ in the melt, as water in the melt, as water in the solid, or in the fluid. The *hydrogen demand* $D^{\mathrm{Hyd}}(y)$ is what all four hold at a trial $y = \ln f_{\mathrm{H_2}}$, with x and n held at their trial values. It is set through the water activity, which the melt and solid water read:

$$a_{\mathrm{H_2O}} = \min\Big(1,\ \frac{K_3\, f_{\mathrm{H_2}}\sqrt{f_{\mathrm{O_2}}}}{\varphi^{\circ}_{\mathrm{H_2O}} P}\Big) \tag{18}$$

the fugacity of water in equilibrium with $f_{\mathrm{H_2}}$ (reaction 3 of Sec. 2) relative to pure water, whose fugacity coefficient $\varphi^{\circ}_{\mathrm{H_2O}}$ comes from Sec. 6.

$$D^{\mathrm{Hyd}}(y) = \underbrace{2\, n^{m}_{\mathrm{H_2}}}_{\text{melt H}_2} + \underbrace{\frac{2\cdot 10^{-2} M_m \big(S_{\mathrm{H_2O}} + S_{\mathrm{OH}}\big)}{m_{\mathrm{H_2O}}}}_{\text{melt water}} + \underbrace{\frac{2\cdot 10^{-2} M_s\, C_s(a_{\mathrm{H_2O}})}{m_{\mathrm{H_2O}}}}_{\text{solid}} + \underbrace{n\big(2X_{\mathrm{H_2O}} + 4X_{\mathrm{CH_4}} + 2X_{\mathrm{H_2}}\big)}_{\text{fluid}}, \qquad F^{\mathrm{Hyd}} = \ln\frac{D^{\mathrm{Hyd}}}{N_{\mathrm{H}}} = 0 \tag{19}$$

M_m and M_s are the melt and solid masses of the cell; the factor $2\cdot 10^{-2}/m_{\mathrm{H_2O}}$ turns wt% of water into moles of H.

Melt $\mathbf{H_2}$ (Hirschmann et al. 2012), with the softcap of 3a and a ceiling $X^{\mathrm{H_2}}_{\max} = 0.3$:

$$n^{m}_{\mathrm{H_2}} = n_{\mathrm{ox}}\frac{X^{m}_{\mathrm{H_2}}}{1 - X^{m}_{\mathrm{H_2}}}, \qquad X^{m}_{\mathrm{H_2}} = \mathrm{softcap}\big(v_h,\, X^{\mathrm{H_2}}_{\max}\big), \qquad v_h = K_{\mathrm{H_2}} f_{\mathrm{H_2}}, \qquad K_{\mathrm{H_2}} = \exp\Big[\ln K_0 - \frac{0.1\,\Delta V\,(P-1)}{RT}\Big] \tag{20}$$

Melt water (Sun & Yao 2026): molecular water $S_{\mathrm{H_2O}}$ and hydroxyl S_{OH} in wt%, read at the water pressure $P_{\mathrm{H_2O}} = a_{\mathrm{H_2O}} P_c$, with P_c and T' the pressure and temperature clamped to the law's calibration and $a_1 \ldots a_{10}$ its coefficients:

$$\ln S_{\mathrm{H_2O}} = a_1 \ln P_{\mathrm{H_2O}} + a_2 \ln T' + \frac{a_3 P_c^2}{T'} + \frac{(a_4\sqrt{P_c} + a_5) X_{\mathrm{Na_2O}}}{T'}, \qquad \ln S_{\mathrm{OH}} = a_6 \ln P_{\mathrm{H_2O}} + a_7 \ln T' + \frac{(a_8 + a_9 X_{\mathrm{Al_2O_3}} + a_{10} X_{\mathrm{Na_2O}}) P_c}{T'}$$
$$P_c = \min(P,\, 2\times 10^4 \text{ bar}), \qquad T' = \max(T,\, 933.15 \text{ K}) \tag{21}$$

Solid water. $C_s(a)$ is the solid's water capacity at activity a (wt%), read from the table at the cell's P, T and composition at 16 activities a_k with values C_k, joined by straight lines in $\ln C$–$\ln a$. Above $a = 1$ it is $C_s(1)$, below the lowest node the last slope continues.

Fluid. The hydrogen species of Sec. 2.

4b. Hydrogen: the derivative

Two slopes first. Below its cap $a_{\mathrm{H_2O}} \propto f_{\mathrm{H_2}}$, so $d\ln a_{\mathrm{H_2O}}/dy = 1$, and 0 at $a_{\mathrm{H_2O}} = 1$. Between nodes k and $k+1$ the solid curve is a power law, so its log-slope is the segment slope:

$$s_k = \frac{d\ln C_s}{d\ln a} = \frac{\ln C_{k+1} - \ln C_k}{\ln a_{k+1} - \ln a_k}, \qquad s_k = 0 \text{ at } a \geq 1 \tag{22}$$

The melt water law is two power laws in $P_{\mathrm{H_2O}}$, so its log-slopes are a_1 and a_6. The melt H_2 uses the softcap slope σ of 3b with v_h and $X^{\mathrm{H_2}}_{\max}$.
The derivative. As for carbon, the solve differentiates $F^{\mathrm{Hyd}} = \ln D^{\mathrm{Hyd}} - \ln N_{\mathrm{H}}$. The terms of $\partial D^{\mathrm{Hyd}}/\partial y$ are, in order, melt H_2, melt water and solid (both carrying $d\ln a_{\mathrm{H_2O}}/dy$), and fluid:

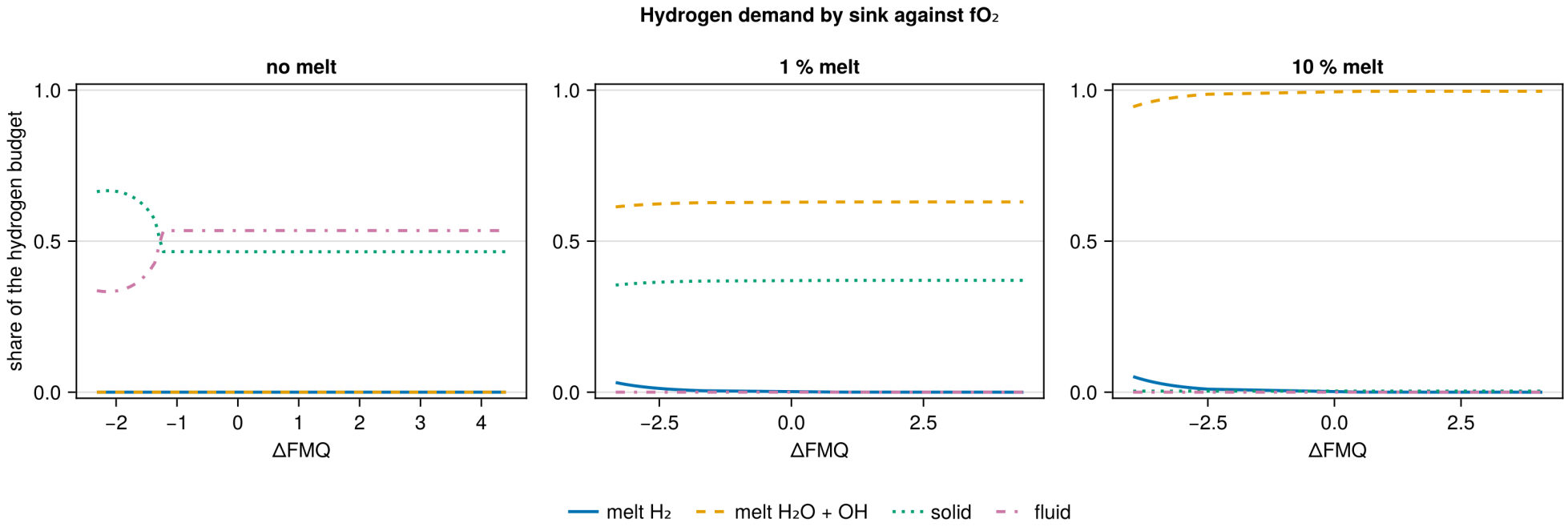
$$\frac{\partial F^{\mathrm{Hyd}}}{\partial y} = \frac{1}{D^{\mathrm{Hyd}}}\Big[\frac{2n_{\mathrm{ox}}\sigma(v_h)v_h}{(1-X^m_{\mathrm{H_2}})^2} + \frac{2\cdot 10^{-2}}{m_{\mathrm{H_2O}}}\big(M_m(a_1 S_{\mathrm{H_2O}} + a_6 S_{\mathrm{OH}}) + M_s C_s\, s_k\big) + n\big(2X_{\mathrm{H_2O}} + 8X_{\mathrm{CH_4}} + 2X_{\mathrm{H_2}}\big)\Big] \tag{23}$$

The couplings to carbon (only through CH_4) and to the fluid amount are

$$\frac{\partial F^{\mathrm{Hyd}}}{\partial x} = \frac{4nX_{\mathrm{CH_4}}}{D^{\mathrm{Hyd}}}, \qquad \frac{\partial F^{\mathrm{Hyd}}}{\partial u} = \frac{n\big(2X_{\mathrm{H_2O}} + 4X_{\mathrm{CH_4}} + 2X_{\mathrm{H_2}}\big)}{D^{\mathrm{Hyd}}} \tag{24}$$

Jumps at the nodes. s_k is exact for the interpolant but jumps at each of the 16 nodes. Newton and Illinois agree there to 3×10^{-8} in $a_{\mathrm{H_2O}}$ (node test of 2026-11-09).
The hydrogen floor. D^{Hyd} rises with y. If it exceeds N_{H} even at the lowest y, the solid alone holds more than the budget. Without melt the solid then takes the whole budget; with melt the cell fails. Bracket of y: $y_{\mathrm{ref}} - 140$ to $y_{\mathrm{ref}} + 20$, with y_{ref} the $\ln f_{\mathrm{H_2}}$ of a carbon-free H_2O–H_2 fluid whose fractions sum to one. Tolerance $|F^{\mathrm{Hyd}}| \leq 10^{-12}$.

4c. Hydrogen: demand by sink against fO_2



4d. Hydrogen: variables

variable	what it is
$y = \ln f_{\mathrm{H_2}}$	log of the H_2 fugacity, the unknown
$D^{\mathrm{Hyd}}(y)$	hydrogen demand: moles of H all sinks hold at y
F^{Hyd}	hydrogen residual, $\ln(D^{\mathrm{Hyd}}/N_{\mathrm{H}})$
N_{H}	hydrogen budget of the cell, moles of H (tracers)
x, $u = \ln n$	carbon potential and fluid amount, at their trial values
$a_{\mathrm{H_2O}}$	water activity, relative to pure water
K_3	equilibrium constant of $H_2 + \frac{1}{2}O_2 = H_2O$
$\varphi^{\circ}_{\mathrm{H_2O}}$	fugacity coefficient of pure water (Sec. 6)
M_m, M_s	melt and solid mass of the cell
$m_{\mathrm{H_2O}}$	molar mass of water
$n^m_{\mathrm{H_2}}$, $X^m_{\mathrm{H_2}}$	moles and mole fraction of H_2 in the melt
$K_{\mathrm{H_2}}$, $\ln K_0$, ΔV	H_2 solubility constant of the melt, its 1-bar value and volume (Hirschmann et al. 2012)

variable	what it is
$X^{\mathrm{H_2}}_{\mathrm{max}}$	ceiling of the melt H_2
v_h, $\sigma(v_h)$	melt H_2 before its ceiling, and the softcap slope
$S_{\mathrm{H_2O}}$, S_{OH}	molecular water and hydroxyl in the melt, wt%
$a_1 \ldots a_{10}$	coefficients of the melt water law (Sun & Yao 2026)
$P_{\mathrm{H_2O}}$	water pressure the melt sees, $a_{\mathrm{H_2O}} P_c$
P_c, T'	P and T clamped to the law's calibration
$X_{\mathrm{Na_2O}}$, $X_{\mathrm{Al_2O_3}}$	oxide fractions of the basaltic melt
$C_s(a)$	solid water capacity at activity a (StagH2Oa table)
C_k, a_k	its 16 table nodes
s_k	slope of $\ln C_s$ against $\ln a$ between nodes
y_{ref}	$\ln f_{\mathrm{H_2}}$ of a carbon-free H_2O–H_2 fluid summing to one

5a. Solver structure

One cell, from outside in. Each level holds the levels below it fixed while it moves its own unknown.

Outer: fO$_2$ $F^{\mathrm{Oxy}}(f) = 0$ (Sec. 1). Bracketed Illinois on $[-120, 40]$; no derivative, because F^{Oxy} jumps where MAGEMin's assemblage changes. (Fortran: a secant from the cell's stored fO$_2$ and slope first, the brackets as fallback.)

Ideality: φ fixed point (Sec. 6e): solve below at frozen φ, recompute φ from the fluid EoS, repeat until $\max|\Delta\varphi/\varphi| < 10^{-8}$ (at least 2 passes, damped after 4).

Fluid exists? solve carbon and hydrogen at $n = 0$; if $s_X^{\mathrm{Flu}} \le 1$, $n = 0$ and stop here.

Fluid amount: $u = \ln n$ Newton on $F^{\mathrm{Flu}}(u) = \ln s_X^{\mathrm{Flu}} = 0$ (Sec. 2), slope from the implicit derivative of the two closures; any step leaving the bracket is replaced by bisection.

Carbon and hydrogen: (x, y) 2D Newton on $(F^{\mathrm{Car}}, F^{\mathrm{Hyd}}) = 0$ (Secs. 3, 4) at the trial n. Steps capped at 2 log units, projected onto the brackets, halved until $(F^{\mathrm{Car}})^2 + (F^{\mathrm{Hyd}})^2$ drops (Armijo). Starts from the cell's last roots.

Fallback: carbon closed by a 1D bracketed Newton at every trial y, hydrogen by a 1D bracketed Newton on its total derivative. It also detects the hydrogen floor.

5b. The Jacobians

Inner 2D Newton on $\mathbf{z} = (x, y)$ at fixed n, from Secs. 3b and 4b:

$$\mathbf{J} = \frac{\partial(F^{\mathrm{Car}}, F^{\mathrm{Hyd}})}{\partial(x, y)} = \begin{pmatrix} \dfrac{\partial F^{\mathrm{Car}}}{\partial x} & \dfrac{\partial F^{\mathrm{Car}}}{\partial y} \\ \dfrac{\partial F^{\mathrm{Hyd}}}{\partial x} & \dfrac{\partial F^{\mathrm{Hyd}}}{\partial y} \end{pmatrix} = \begin{pmatrix} \dfrac{1}{D^{\mathrm{Car}}}\dfrac{\partial D^{\mathrm{Car}}}{\partial x} & \dfrac{2nX_{\mathrm{CH_4}}}{D^{\mathrm{Car}}} \\ \dfrac{4nX_{\mathrm{CH_4}}}{D^{\mathrm{Hyd}}} & \dfrac{1}{D^{\mathrm{Hyd}}}\dfrac{\partial D^{\mathrm{Hyd}}}{\partial y} \end{pmatrix} \tag{25}$$

$$\mathbf{z}_{k+1} = \mathbf{z}_k - \lambda\, \mathbf{J}^{-1}\mathbf{F}(\mathbf{z}_k) \tag{26}$$

A zero budget. If $N_{\mathrm{C}} = 0$ (or $N_{\mathrm{H}} = 0$) its potential is frozen at $-\infty$ and its row is replaced by the identity, $F^{\mathrm{Car}} = x$, so the same 2×2 system covers water-only, carbon-only and mixed cells.

Outer Newton on u uses the augmented 3×3 Jacobian of $(F^{\mathrm{Car}}, F^{\mathrm{Hyd}}, F^{\mathrm{Flu}})$ in (x, y, u):

$$\mathbf{J}_3 = \begin{pmatrix} & \mathbf{J} & & \partial_u F^{\mathrm{Car}} \\ & & & \partial_u F^{\mathrm{Hyd}} \\ \partial_x F^{\mathrm{Flu}} & \partial_y F^{\mathrm{Flu}} & & 0 \end{pmatrix}, \qquad \frac{dF^{\mathrm{Flu}}}{du} = -\begin{pmatrix} \partial_x F^{\mathrm{Flu}}, & \partial_y F^{\mathrm{Flu}} \end{pmatrix} \mathbf{J}^{-1} \begin{pmatrix} \partial_u F^{\mathrm{Car}} \\ \partial_u F^{\mathrm{Hyd}} \end{pmatrix},$$

$$u_{k+1} = u_k - \frac{F^{\mathrm{Flu}}(u_k)}{dF^{\mathrm{Flu}}/du} \tag{27}$$

with $\partial_u F^{\mathrm{Car}} = n(X_{\mathrm{CO_2}} + X_{\mathrm{CO}} + X_{\mathrm{CH_4}})/D^{\mathrm{Car}}$, $\partial_u F^{\mathrm{Hyd}} = n(2X_{\mathrm{H_2O}} + 4X_{\mathrm{CH_4}} + 2X_{\mathrm{H_2}})/D^{\mathrm{Hyd}}$, $\partial_x F^{\mathrm{Flu}} = (X_{\mathrm{CO_2}} + X_{\mathrm{CO}} + X_{\mathrm{CH_4}})/s_X^{\mathrm{Flu}}$, $\partial_y F^{\mathrm{Flu}} = (X_{\mathrm{H_2O}} + 2X_{\mathrm{CH_4}} + X_{\mathrm{H_2}})/s_X^{\mathrm{Flu}}$.

How they are built. One ForwardDiff pass in (x, y) gives $\mathbf{F}$ and $\mathbf{J}$; one pass in (x, y, u) gives $\mathbf{J}_3$. Nothing above is coded by hand; the expressions are what those passes return, and they hold the table slope s_k of Sec. 4b.

5c. Solver: variables

variable	what it is
$\mathbf{z} = (x, y)$	potentials of carbon and hydrogen, iterates of the inner Newton
$u = \ln n$	log of the fluid moles, iterate of the outer Newton
$\mathbf{F} = (F^{\mathrm{Car}}, F^{\mathrm{Hyd}})$	carbon and hydrogen residuals (Secs. 3, 4)
$F^{\mathrm{Flu}}(u)$	fluid closure residual (Sec. 2)
$\mathbf{J}$	Jacobian of $\mathbf{F}$ in (x, y)
$\mathbf{J}_3$	Jacobian of $(F^{\mathrm{Car}}, F^{\mathrm{Hyd}}, F^{\mathrm{Flu}})$ in (x, y, u)
λ	line-search factor, halved until $(F^{\mathrm{Car}})^2 + (F^{\mathrm{Hyd}})^2$ drops (Armijo)

variable	what it is
$\partial_u F^{\mathrm{Car}}$, $\partial_u F^{\mathrm{Hyd}}$	how the closures shift with the fluid amount
$\partial_x F^{\mathrm{Flu}}$, $\partial_y F^{\mathrm{Flu}}$	how the fluid closure shifts with the potentials
s_k	table slope of the solid water curve (Sec. 4b)
$X_{\mathrm{CH_4}}$	methane fraction, the only coupling of x and y
N_{C}, N_{H}	carbon and hydrogen budgets; a zero one freezes its row

Settings: steps capped at 2 log units, up to 8 halvings (Armijo 10^{-4}), at most 40 inner and 100 outer iterations, tolerances $|F^{\mathrm{Car}}|, |F^{\mathrm{Hyd}}|, |F^{\mathrm{Flu}}| \le 10^{-12}$, closure check 10^{-6} relative. Seeds: the cell's last (x, y, u) and φ.

6a. Fluid ideality I: ZD09, the volume (unknown V_m; Zhang & Duan 2009, used at $T \geq 773$ K)

Secs. 2–4 need one fugacity coefficient $\varphi_i = f_i/(X_i P)$ per species. Above 773 K it comes from ZD09, a single equation of state fitted to *methane*. Every other species and every mixture is mapped onto methane by corresponding states: temperature in units of the attraction energy ε, pressure in units of ε/σ^3. The hybrid of 6d–6e covers 673 K and below, with a blend in between.

Step 1, mix the molecular parameters (attraction as a geometric mean, size as an arithmetic mean):

$$\varepsilon = \sum_{ij} x_i x_j k_{ij}^{(1)} \sqrt{\varepsilon_i \varepsilon_j}, \qquad \sigma = \sum_{ij} x_i x_j k_{ij}^{(2)} \tfrac{\sigma_i+\sigma_j}{2}, \qquad \varepsilon_i' = \sum_{j} x_j k_{ij}^{(1)} \sqrt{\varepsilon_i \varepsilon_j}, \qquad \sigma_i' = \sum_{j} x_j k_{ij}^{(2)} \tfrac{\sigma_i+\sigma_j}{2} \tag{28}$$

Step 2, scale onto methane ($\varepsilon_{\mathrm{CH_4}} = 154$ K, $\sigma_{\mathrm{CH_4}} = 3.691$ Å):

$$T_m = T\,\frac{154}{\varepsilon}, \qquad P_m = P\left(\frac{\sigma}{3.691}\right)^3 \frac{154}{\varepsilon} \tag{29}$$

Step 3, solve methane's equation for its molar volume (L/mol), a virial-like series in $1/V_m$ plus one exponential term for dense fluids:

$$Z = \frac{P_m V_m}{R T_m} = 1 + \frac{g_1}{V_m} + \frac{g_2}{V_m^2} + \frac{g_3}{V_m^4} + \frac{g_4}{V_m^5} + \frac{c_{13}}{T_m^3 V_m^2}\left(c_{14} + \frac{c_{15}}{V_m^2}\right) e^{-c_{15}/V_m^2}, \qquad g_k = c_{3k-2} + \frac{c_{3k-1}}{T_m^2} + \frac{c_{3k}}{T_m^3} \tag{30}$$

Derivative. Newton runs in density $d = 1/V_m$ on the pressure residual, with its analytic slope ($g_5 = c_{13}/T_m^3$, $q = c_{14} + c_{15} d^2$, $e = e^{-c_{15} d^2}$):

$$F(d) = R\,T_m\,d\,Z(d) - P_m = 0, \qquad d_{k+1} = d_k - \frac{F}{dF/dd}$$
$$\frac{dF}{dd} = R\,T_m\Big[1 + 2g_1 d + 3g_2 d^2 + 5g_3 d^4 + 6g_4 d^5 + g_5\, e\, d^2\Big(3q + 2c_{15} d^2 (1-q)\Big)\Big] \tag{31}$$

Brackets in V_m of $[1, 1000]$, $[0.01, 10]$ and $[0.001, 1]$ are tried in turn, widest first, so the fluid-like root is taken; a step leaving the bracket is replaced by bisection. The real volume scales back with the molecule size, $V = 10^3 V_m (\sigma/3.691)^3$ cm^3/mol.

6b. ZD09: the fugacity coefficient (result φ_i)

Step 4. The coefficient of each species is the mixture's own non-ideality plus two corrections for how i differs from the average molecule, at the volume V_m of 6a:

$$\ln\varphi_i = \underbrace{Z-1-\ln Z+S_1}_{\text{same for every species}} \quad \underbrace{-2S_2\Big(1-\frac{\varepsilon_i'}{\varepsilon}\Big)}_{\text{attraction of } i \text{ vs average}} \quad \underbrace{+6(1-Z)\Big(1-\frac{\sigma_i'}{\sigma}\Big)}_{\text{size of } i \text{ vs average}} \tag{32}$$

$$S_1 = \sum_k \frac{g_k}{p_k V_m^{p_k}} + \frac{c_{13}}{2c_{15}T_m^3}\Big[c_{14}+1-\Big(c_{14}+1+\frac{c_{15}}{V_m^2}\Big)e^{-c_{15}/V_m^2}\Big], \qquad p=(1,2,4,5) \tag{33}$$

$$S_2 = \sum_k \frac{h_k}{p_k V_m^{p_k}} + \frac{3c_{13}}{2c_{15}T_m^3}\Big[c_{14}+1-\Big(c_{14}+1-\frac{c_{15}}{V_m^2}\Big)e^{-c_{15}/V_m^2}\Big], \qquad h_k=\frac{2c_{3k-1}}{T_m^2}+\frac{3c_{3k}}{T_m^3} \tag{34}$$

S_1 is the series of $Z-1$ integrated over volume, term by term. S_2 is the same integral of $-T_m\,\partial(Z-1)/\partial T_m$, which is how strongly the non-ideality depends on attraction. ε_i' and σ_i' are how strongly, and at what distance, species i interacts with the *average* neighbour of this mixture.

$Z = 4.2$ in the example means the fluid fills 4.2 times the volume of an ideal gas at the same P and T: at 3 GPa repulsion dominates. CO_2 is larger and less attractive than the average molecule of this water-rich fluid, so it pays far more for its place (φ 798 against 12.9). That raises its fugacity and pushes carbon out of the fluid.

Why ZD09 is not used cold. It has no liquid-water root below about 600 K. Pure water at 0.3 GPa and 500 K: ZD09 returns $V = 4.5$ cm^3/mol and $\varphi = 6.5\times10^{-28}$, the hybrid of 6d–6e returns $V = 18.3$ cm^3/mol and $\varphi = 0.032$.

Worked example: $X_{H_2O} = 0.9$, $X_{CO_2} = 0.1$, 3 GPa, 1600 K.

Step 1: ε, σ	468.4 K, 2.983 Å
Step 2: T_m, P_m	526.0, 5206
Step 3: V_m	0.03558 L/mol
g_1/V_m, g_2/V_m^2	0.133, 1.020
g_3/V_m^4, g_4/V_m^5, exp	2.191, −0.108, 0
Z	4.235
V	18.78 cm^3/mol
Step 4: $Z-1-\ln Z$, S_1, S_2	1.792, 1.178, −0.957

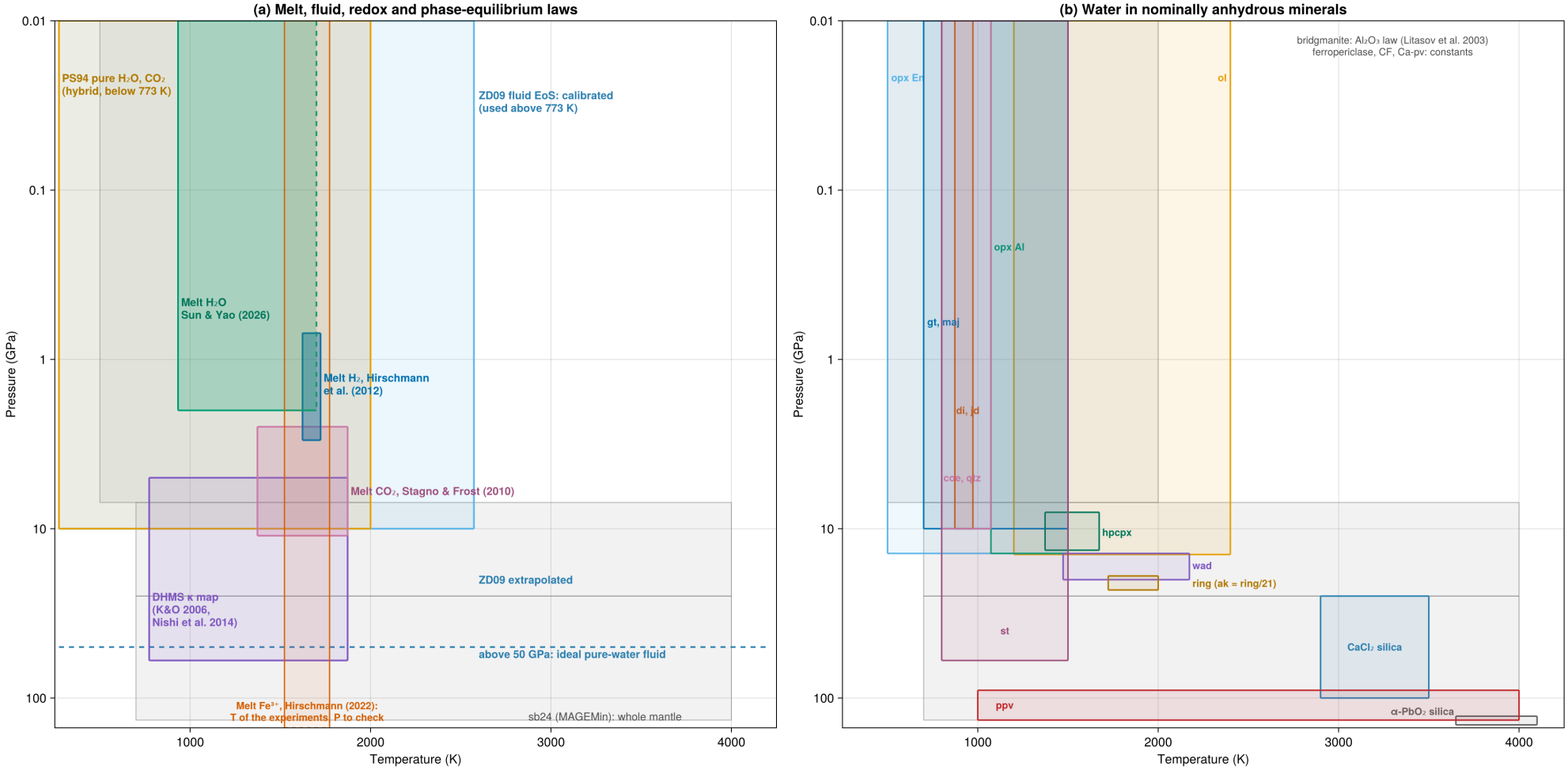
	H_2O	CO_2
$\varepsilon_i'/\varepsilon$	1.043	0.616
σ_i'/σ	0.983	1.153
attraction term	−0.082	+0.736
size term	−0.331	+2.977
$\ln\varphi_i$	2.558	6.683
φ_i	12.9	798

6c. ZD09: variables

variable	what it is
$\varphi_i = f_i/(X_i P)$	fugacity coefficient of species i, the result
V_m	molar volume of the equivalent methane, L/mol, the unknown
V	molar volume of the real fluid, cm^3/mol
x_i	mole fraction of species i (the X_i of Sec. 2)
ε_i, σ_i	depth of the Lennard-Jones well (K) and molecular diameter (Å) of i
$k_{ij}^{(1)}$, $k_{ij}^{(2)}$	corrections for unlike pairs, attraction and size
ε, σ	mixture averages of ε_i and σ_i
ε_i', σ_i'	attraction and distance of i to the average neighbour

variable	what it is
T_m, P_m	temperature and pressure scaled onto methane
Z	compressibility factor, $P_m V_m/(RT_m)$
$c_1 \dots c_{15}$	ZD09 constants of methane
g_k, h_k	temperature-dependent virial coefficients and their attraction part
p_k	powers of $1/V_m$ in the series, (1, 2, 4, 5)
S_1, S_2	volume integrals of $Z-1$ and of its temperature derivative
P, T	pressure and temperature of the cell

7. Calibration ranges of the model laws



Grey: the model's table sectors (upper mantle 0–7 GPa, transition zone 7–25 GPa, lower mantle 25–135 GPa). Dashed edges: a bound not yet checked against the source.

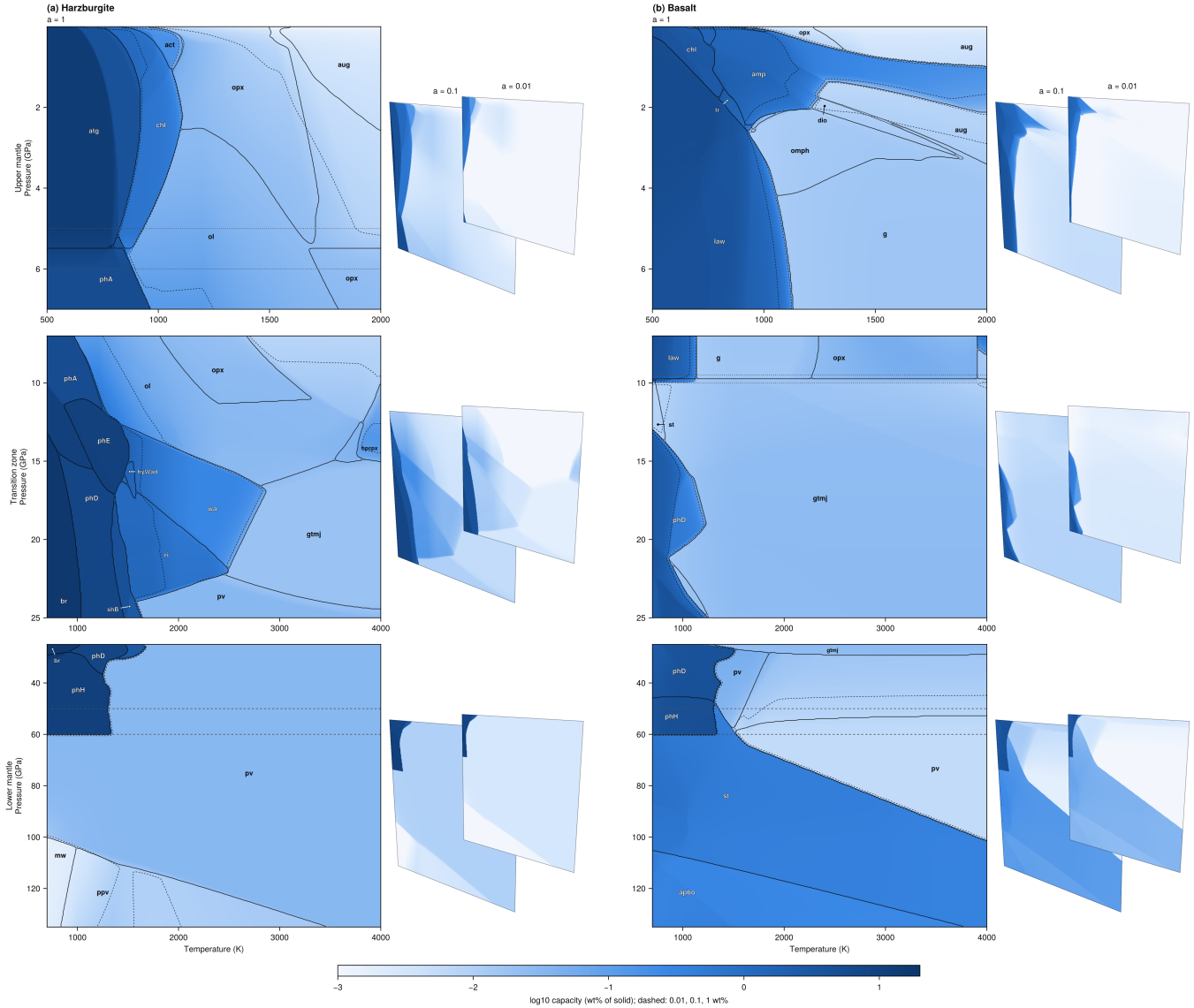


Figure 1: Storage capacity in the three mantle sectors, harzburgite (a) and basalt (b), flat at $a = 1$ with $a = 0.1$ and 0.01 receding (Water paper, Figure 1).

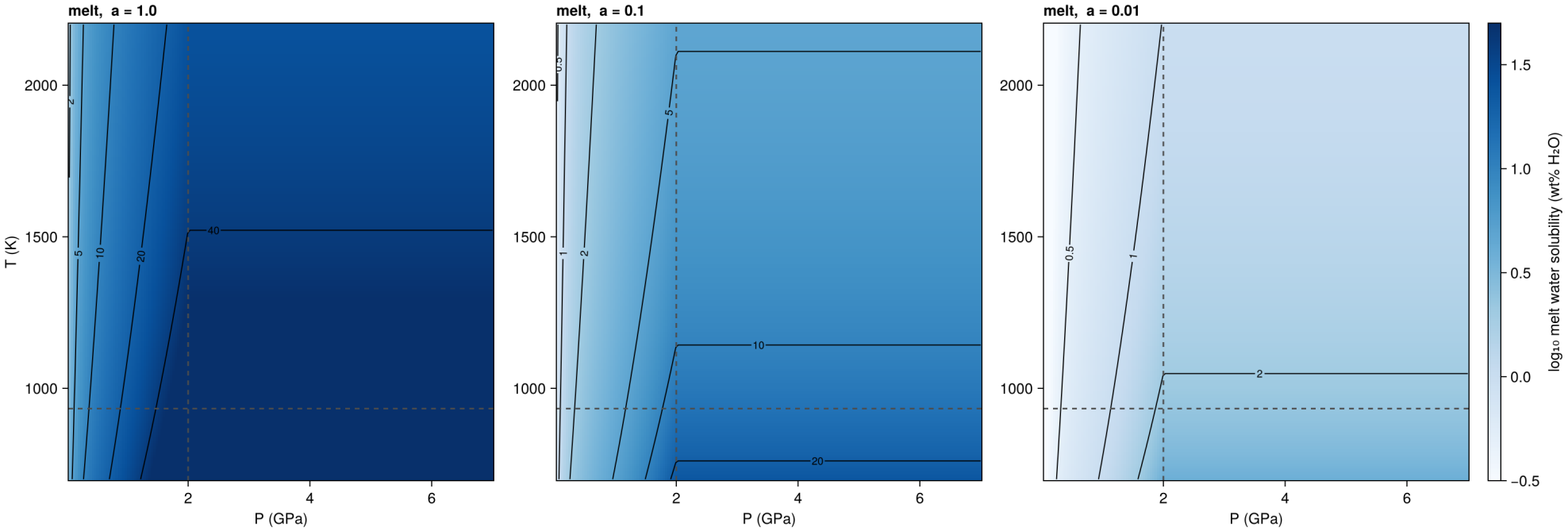
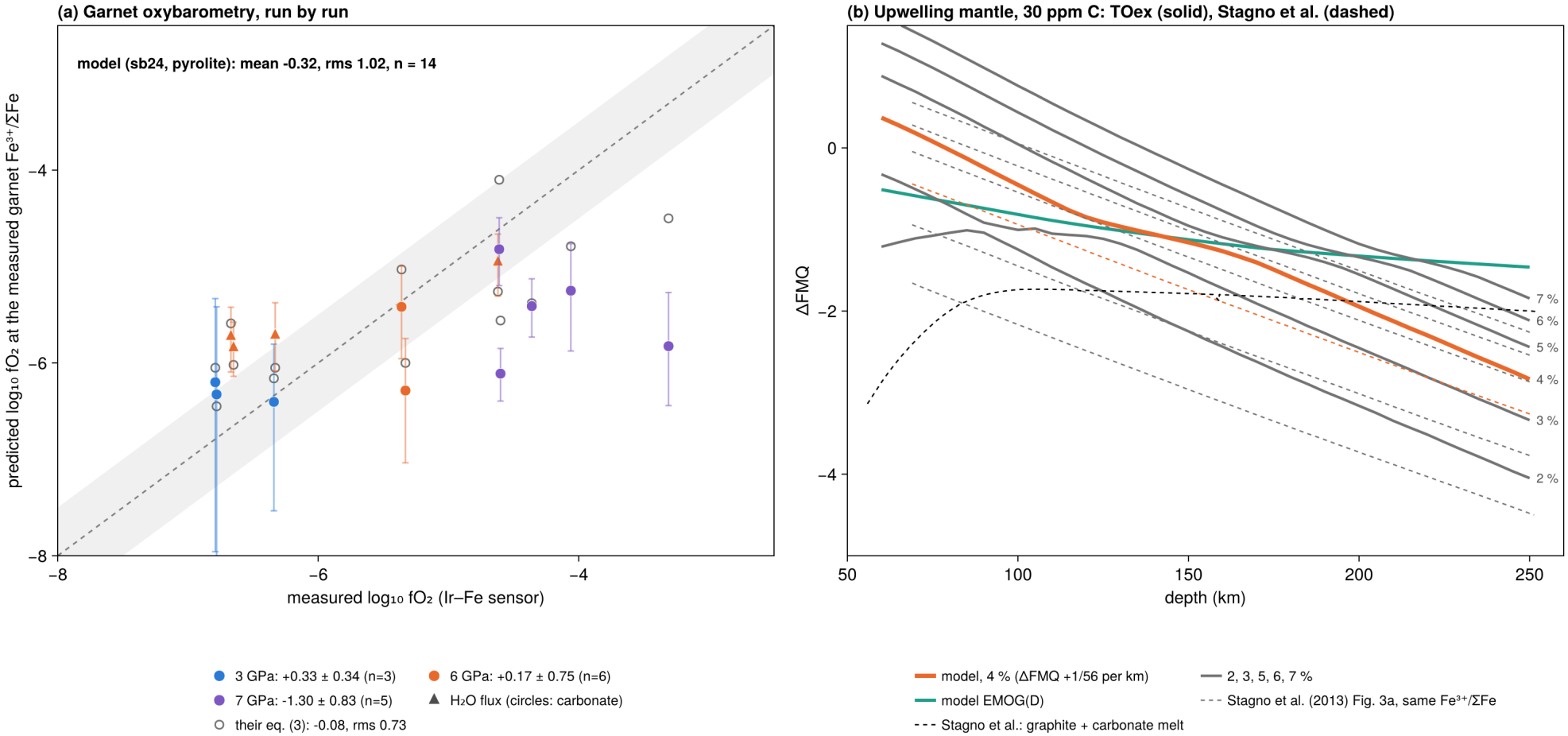
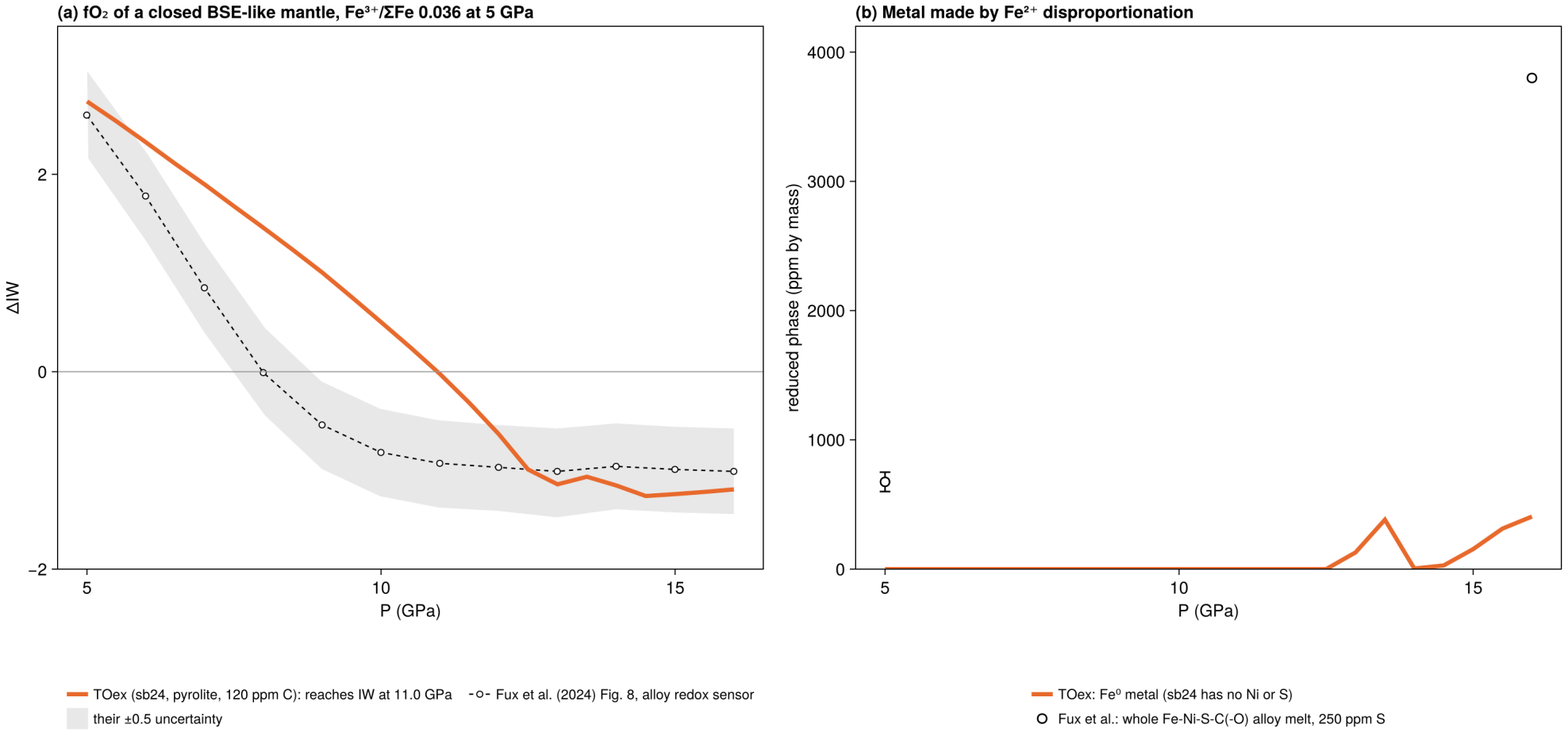


Figure 2: Melt water solubility of the Sun & Yao law read at $P_{\mathrm{H_2O}} = aP$, with P held at 2 GPa and T floored at 933 K.

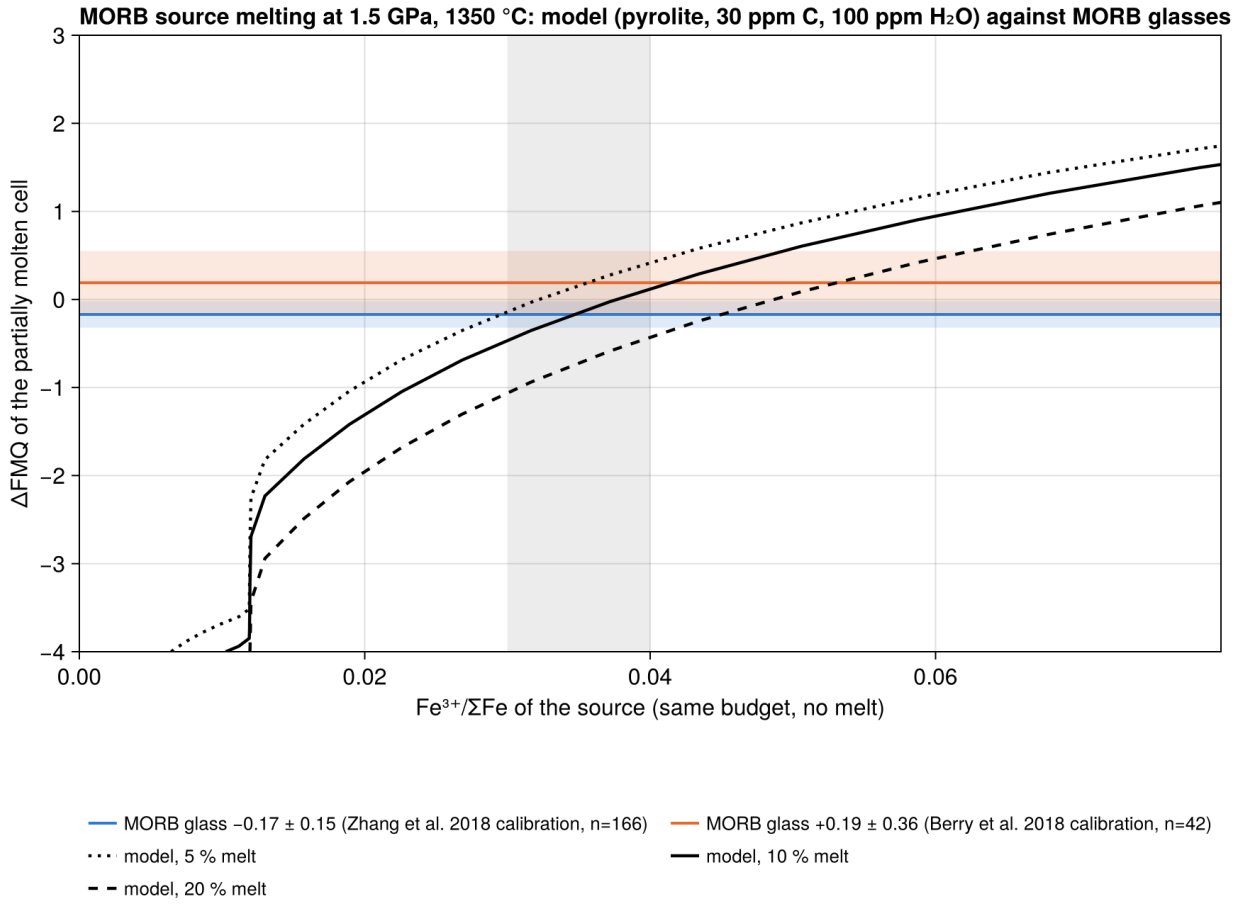
8a. Benchmark against experiments: Stagno et al. (2013)



8b. Benchmark against experiments: Fux et al. (2024)



8c. Benchmark against natural rocks: Cottrell et al. (2021)



grey: fertile-mantle $Fe^{3+}/\Sigma Fe$ 0.03–0.04 (Canil & O'Neill 1996). MORB values: Cottrell et al. (2021) Table 3.1, relative to QFM at 1 atm; the model's FMQ is computed from sb24.

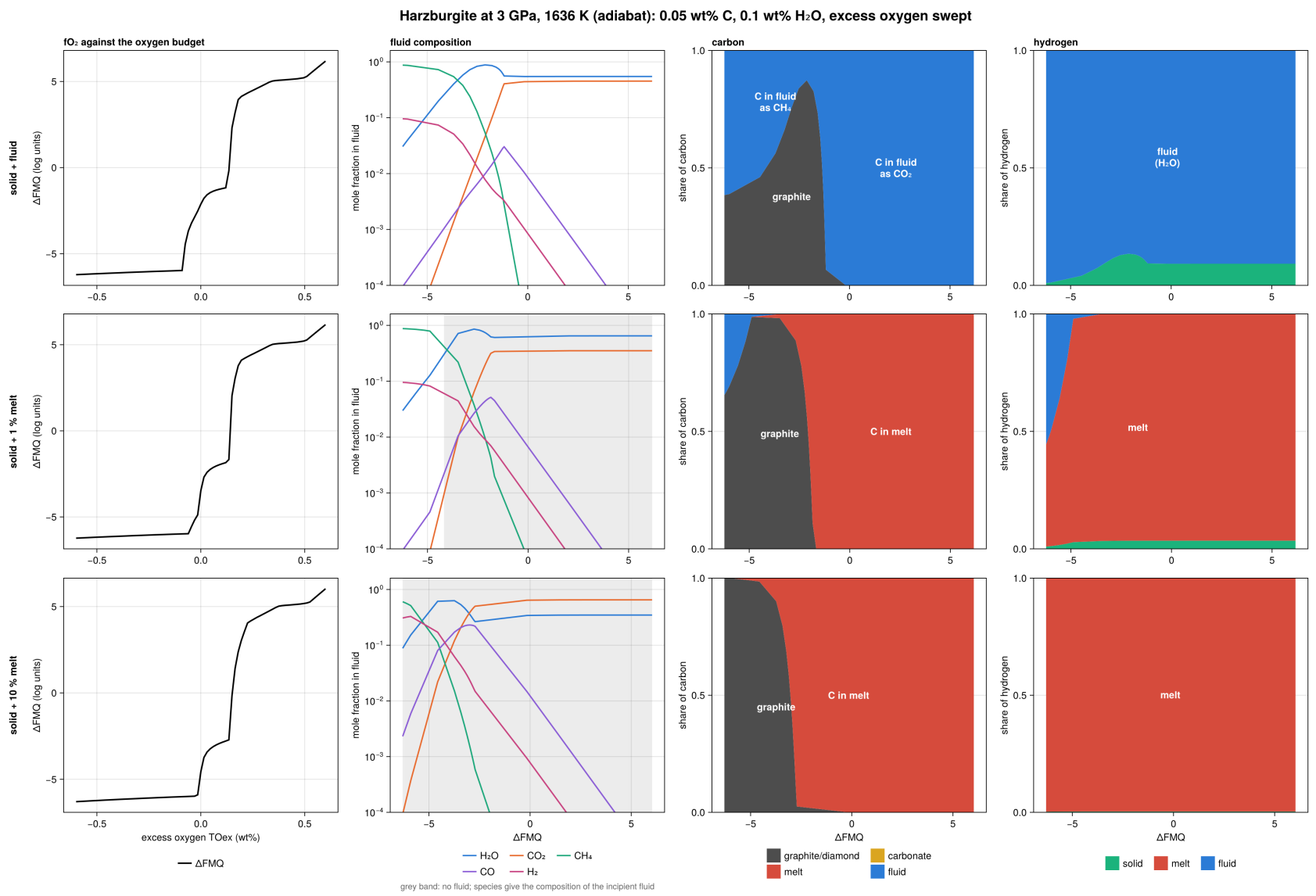


Figure 3: Oxygen budget swept at 3 GPa. Rows: no melt, 1 % and 10 % melt. Column 1 maps the budget to fO_2. Columns 2–4 are plotted against the resulting ΔFMQ.

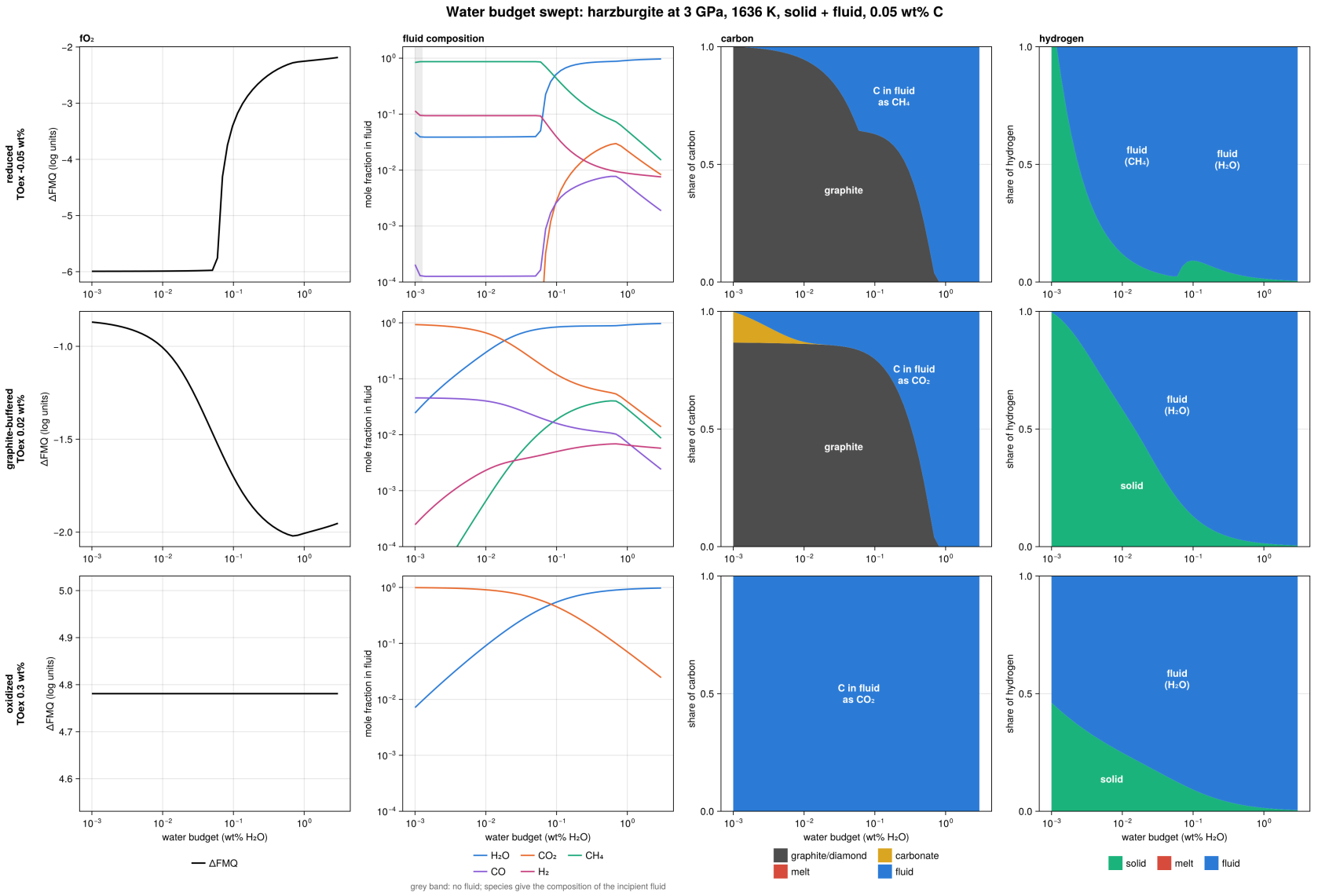


Figure 4: Water budget swept at 3 GPa, no melt. Rows: reduced, graphite-buffered and oxidized oxygen budgets.

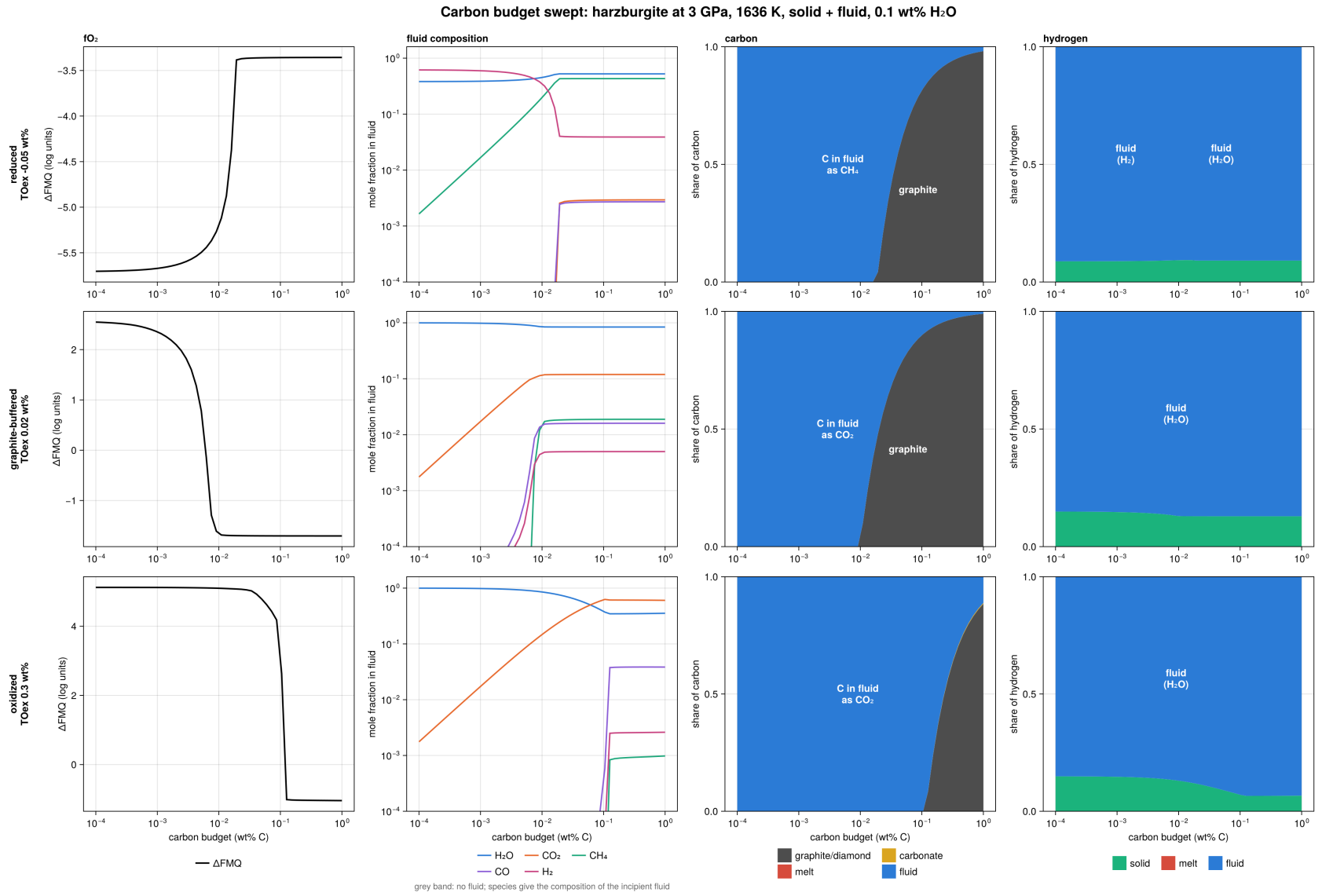


Figure 5: Carbon budget swept at 3 GPa, no melt. Rows as in Fig. 4.

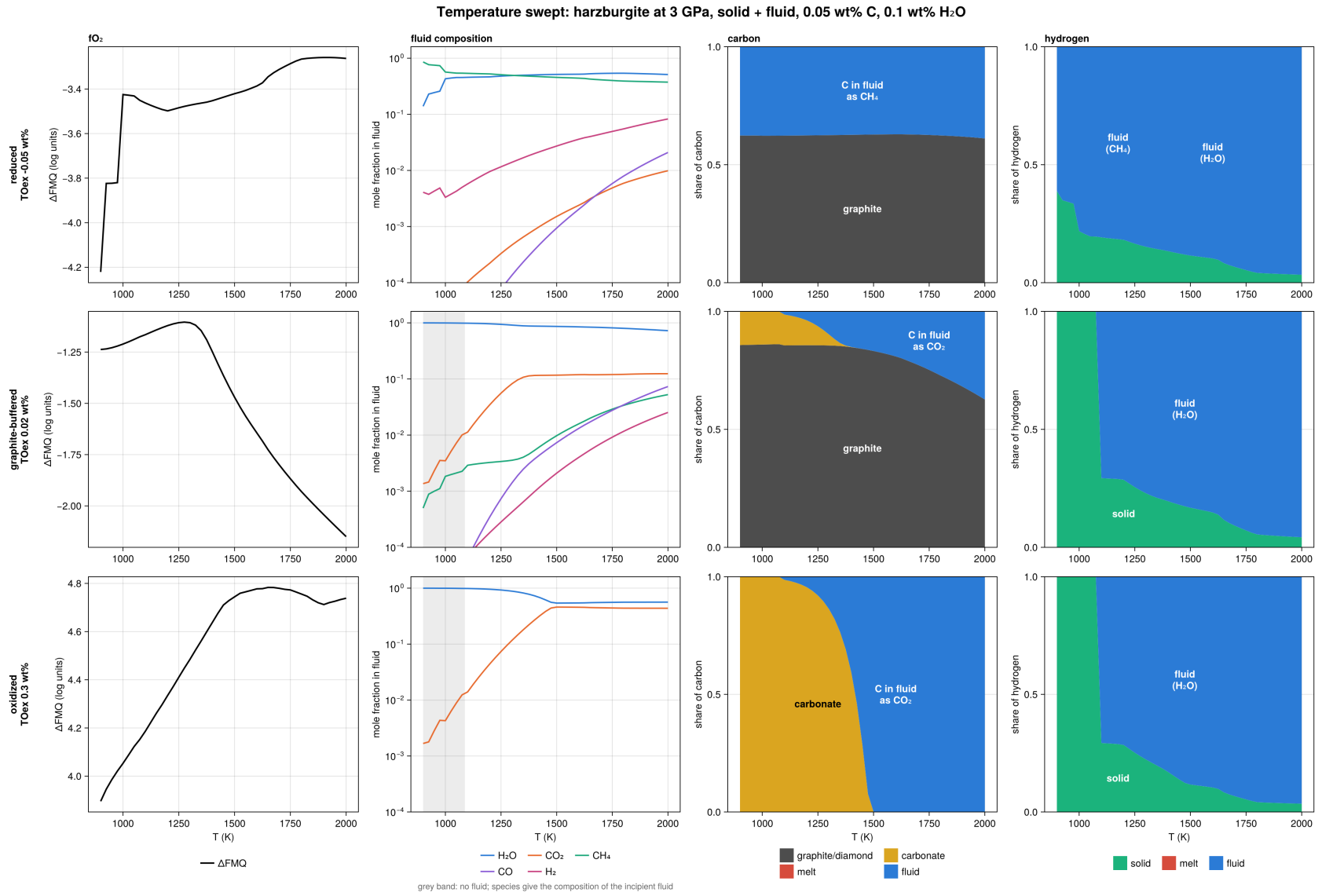


Figure 6: Temperature swept at 3 GPa, no melt. Rows as in Fig. 4.